



Animal Pose Estimation Based on YOLO-POSE

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Abstract. With the development of computer vision technology, deep learning-based pose estimation and target detection have been widely used in the fields of human behavior analysis and intelligent security. However, owing to the complexity of animal poses and the diversity of species, the existing pose estimation methods still face many challenges when applied to animal targets. To solve this problem, an improved YOLO-Pose model is proposed to improve the accuracy and efficiency of animal pose estimation. On the basis of the original YOLO-Pose model, a separable kernel attention mechanism is introduced and improved to make it conform to the animal target, and combined with the spatial pyramid pool of YOLO-Pose, the multiscale feature fusion capability of the model is improved. The experimental results show that the improved YOLO-Pose model achieves excellent performance on both the public animal pose dataset and the AP-10K dataset, significantly improving the ability of target detection and pose estimation.

Keywords: Animal · Pose estimation · LSKA · YOLO-Pose

1 Introduction

As a fundamental task in computer vision, human and animal pose estimation has found widespread applications across intelligent surveillance, human-computer interaction, and motion analysis domains. Traditional approaches predominantly follow a two-stage “detection-then-estimation” paradigm, where an object detector first identifies target instances before a separate pose estimation model predicts keypoints within each detection region. While achieving notable accuracy, this sequential processing introduces significant computational redundancy, often compromising real-time performance requirements in practical applications.

The evolution of pose estimation methods has led to several important developments. Early deep learning approaches such as convolutional pose machines (CPMs) [1] employed sequential convolutional networks to model spatial relationships, whereas

stacked hourglass networks [2] introduced symmetric encoder decoder architectures for multiscale feature fusion. The high-resolution network (HRNet) [3] further advanced the field by maintaining high-resolution feature representations through parallel multi-branch structures. Subsequent innovations have focused on optimizing various aspects of the two-stage paradigm: OKDHP [4] introduced online knowledge distillation to increase model efficiency, OpenPose [5] improved geometric modelling through Part Affinity Fields, and DisPose [6] addressed occlusion challenges via spatial dependency decoupling. Recent breakthroughs in two-stage methods include ViTPose++ [7], which combines vision transformers with deformable convolutions to achieve state-of-the-art performance on benchmark datasets, and DiffPose [8], which leverages diffusion models to refine keypoint localization in low-quality images. Despite these advances, two-stage approaches remain constrained by inherent limitations including computational inefficiency from repeated feature extraction, architectural bottlenecks that impede real-time processing, and limited generalizability to nonhuman targets.

In contrast, single-stage pose estimation methods have emerged as promising alternatives by unifying detection and keypoint prediction within an end-to-end framework. These approaches typically employ multitask learning architectures that share backbone feature extraction for simultaneous bounding box regression and keypoint coordinate prediction, significantly improving computational efficiency. Representative single-stage models include YOLO-Pose [9], RTM-Pose [10], and PETR [11], each offering distinct advantages in speed accuracy trade-offs [12]. Among these, YOLO-based architectures have gained particular prominence, because of their exceptional real-time performance, deployment simplicity, and hardware compatibility, leading to numerous variants [13] such as YOLO-Pose v2 [14] and YOLO-rlepose [15], which continue to push the boundaries of efficient pose estimation.

Despite these significant advancements, current single-stage pose estimation systems still face critical challenges that limit their practical deployment. The persistent difficulty in balancing model lightweightness with prediction accuracy remains a fundamental constraint, particularly for edge computing applications with stringent resource requirements. Furthermore, the predominant focus on human pose estimation in existing research has resulted in limited generalizability to diverse animal species, creating significant barriers for applications in agricultural monitoring, veterinary science, and wildlife conservation. These unresolved challenges underscore the need for continued innovation in developing unified, multispecies pose estimation frameworks that can maintain real-time performance while achieving robust accuracy across various target domains.

To address these challenges, this paper proposes a lightweight animal pose estimation framework based on YOLO-Pose, which is designed for multiscene, multispecies scenarios. By introducing and improving the large separable kernel attention (LSKA) [16] mechanism, the model achieves ultrafast inference while maintaining accuracy, providing a new solution for real-time pose analysis in edge computing. The contributions of this work are as follows:

- An improved large separable kernel attention modules is deeply integrated with the SPPF module. By utilizing large kernel space attention and channel decoupling, the accuracy of key point detection can be improved without adding additional computational overhead.

- Unlike most existing works that focus on a single species or specific scenario, this study targets animal detection across diverse real-world scenes, offering a reference framework for future research.

2 Method

Vision attention networks incorporating large kernel attention (LKA) modules have gained significant attention, because of their impressive performance. However, as the k_size increases in these networks, the memory demands of the deep convolutional layers within the large kernel attention module also increase proportionally. To address this challenge, Kin Wai Lau, Lai-Man Po, and others [16] conducted a series of experiments. Typically, for an input feature map $F \in \mathbb{R}^{C \times H \times W}$, the output of the LKA can be computed via formulas (1), (2), and (3):

$$Z^C = \sum_{H,W} W_{k \times k}^C * F^C \quad (1)$$

$$A^C = W_{1 \times 1} * Z^C \quad (2)$$

$$\bar{F}^C = A^C \otimes F^C \quad (3)$$

In the formula above, C represents the count of input channels, $*$ indicates convolution, and $\otimes$ denotes the Hadamard product. H and W denote the height and width of the feature map, respectively. Z^C refers to the output obtained by convolving the feature map $\bar{F}^C$ with a $k \times k$ kernel, where F^C is the input feature map and where A^C is the attention map.

Next, the depthwise convolution with a large kernel is decomposed into two stages: first, a depthwise convolution using smaller kernels, then a dilated depthwise convolution with a relatively larger kernel. In this case, the output equation of the LKA is given by the following expressions (4), (5), (6), and (7):

$$\bar{Z}^C = \sum_{H,W} W_{(2d-1) \times (2d-1)}^C * F^C \quad (4)$$

$$Z^C = \sum_{H,W} W_{[k/d] \times [k/d]}^C * \bar{Z}^C \quad (5)$$

$$A^C = W_{1 \times 1} * Z^C \quad (6)$$

$$\bar{F}^C = A^C \otimes F^C \quad (7)$$

Here, Z^C denotes the result of a depthwise convolution with a kernel size of $(2d - 1) \times (2d - 1)$, where d represents the dilation rate.. To complement this operation, $\bar{Z}^C$ captures local spatial cues and serves as a corrective mechanism for depthwise dilated convolutions, which utilize an effective kernel size of $[k/d] \times [k/d]$. The dilated depthwise convolution plays a key role in acquiring global spatial context from the depthwise convolutional output Z^C . Although this approach reduces the computational burden

associated with large convolutional kernels, it still incurs significant computational and memory overhead when the kernel size exceeds 23×23 , posing efficiency challenges for the VAN architecture.

To address this issue, Kin WaiLau, Lai-Man Po, and others [16] proposed an equivalent improved configuration for LKA by decomposing the two-dimensional weight kernel of depthwise convolution into two cascaded one-dimensional separable kernels. This modification of the LKA module is referred to as LSKA. The output of the LSKA is given by Eqs. (8), (9), (10), and (11):

$$\bar{Z}^C = \sum_{H,W} W_{(2d-1) \times 1}^C * (\sum_{H,W} W_{1 \times (2d-1)}^C * F^C) \quad (8)$$

$$Z^C = \sum_{H,W} W_{[k/d] \times 1}^C * (\sum_{H,W} W_{1 \times [k/d]}^C * \bar{Z}^C) \quad (9)$$

$$A^C = W_{1 \times 1} * Z^C \quad (10)$$

$$\bar{F}^C = A^C \otimes F^C \quad (11)$$

The core innovation of the LSKA module lies in the dimensional decoupling of traditional large-kernel convolutions. By decomposing the two-dimensional depthwise convolution kernel into cascaded vertical and horizontal one-dimensional convolution kernels ($K = K_v \otimes K_h$), the computational complexity is optimized from $O(k^2)$ to $O(2k)$, achieving linear optimization. This orthogonal decomposition strategy imposes constraints on the parameter space, reducing the number of parameters to $1/k$ (where k is the kernel size) while maintaining an equivalent receptive field. Specifically, a tensor operation model can be established as follows:

$$LSKA(X) = Conv1D_h(Conv1D_v(X)) + Identity(X) \quad (12)$$

The formula represents a vertical 1D convolution of the input feature map X followed by a horizontal 1D convolution. Finally, the convolution results are added to the original input x , which is often used to enhance feature extraction while preserving the original input information to avoid information loss.

When the kernel size is expanded to 31×31 , the traditional LKA module requires 4.82 GFLOPs, whereas LSKA only requires 0.96 GFLOPs, achieving a $5.02 \times$ improvement in computational efficiency. In terms of memory usage, LSKA reduces peak GPU memory consumption on ImageNet-1K by 62.3% (from 3.2 GB to 1.2 GB). This optimization stems from the rank constraint of the weight matrix—the decomposed dual one-dimensional convolution kernels have a rank of 2, whereas the original two-dimensional kernel has a rank of k^2 , significantly mitigating the dimensionality curse in matrix operations. The improved module introduces a BN-Dropout residual structure, forming a dual regularization mechanism. The batch normalization (BN) layer constrains the feature distribution to $N(0,1)$, alleviating the covariate shift, whereas the random mask regularization of dropout enhances robustness. A progressive dropout strategy is employed, preserving detailed features in shallow layers and enhancing generalization in deeper layers. The residual connection constructs an identity mapping $H(x) = F(x) + x$, ensuring

smooth gradient flow, and the experiments demonstrate that it improves the convergence speed. When integrating LSKA into the SPPF structure, a hierarchical kernel size strategy (7/11/19/23) is adopted, controlling the expansion rate of the receptive field through dilated convolutions. Cross-scale attention gates are established between levels of the feature pyramid, and sigmoid weighting is used to achieve feature selection.

In addition, residual connections enable more diverse information propagation within the network while preserving the details and contextual information of the original input data. This connection method ensures that no critical information is lost during the transmission process, thereby enhancing the model’s performance in subsequent tasks. More importantly, residual connections also help alleviate the vanishing gradient problem, making the training of deep networks more manageable. The architecture of the improved model is illustrated in Fig. 1.

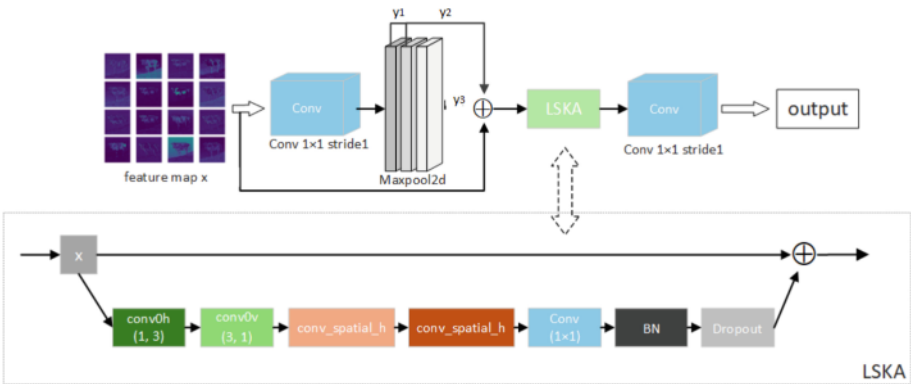


Fig. 1. Top: diagram of SPPF integrated with the LSKA structure; bottom: diagram of the improved LSKA structure.

In the LSKA-SPPF (Spatial Pyramid Pooling - Fast with Local Spatial Kernel Attention) structure, the feature graph x obtained in the backbone is convolved 1×1 . The number of channels is adjusted and the features are preliminarily fused. Then the spatial information of the feature map is downsampled through the maximum pooling layer. Through the three-layer pooling operation, the spatial information of different scales can be captured, and the model can enhance the perception ability of objects of different sizes and output y_1 , y_2 and y_3 . By combining the original feature map x with the feature maps y_1 , y_2 and y_3 after pooling at different scales, a feature representation containing rich multiscale information is formed. The LSKA module is designed to enhance the extraction of critical information from feature maps by emphasizing relevant regions and suppressing irrelevant ones. Through the integration of an attention mechanism, the model is guided to concentrate on feature regions that are most beneficial to the task. Subsequently, the feature map refined by the LSKA module undergoes a 1×1 convolution to adjust the channel dimensions and facilitate feature fusion. This operation aids in projecting the high-level representations obtained by the LSKA module into an output space aligned with the task objectives. The inherent multiscale feature fusion

capability of the SPPF module is further strengthened through the incorporation of the LSKA module. The complete architectural design of the model is illustrated in Fig. 2.

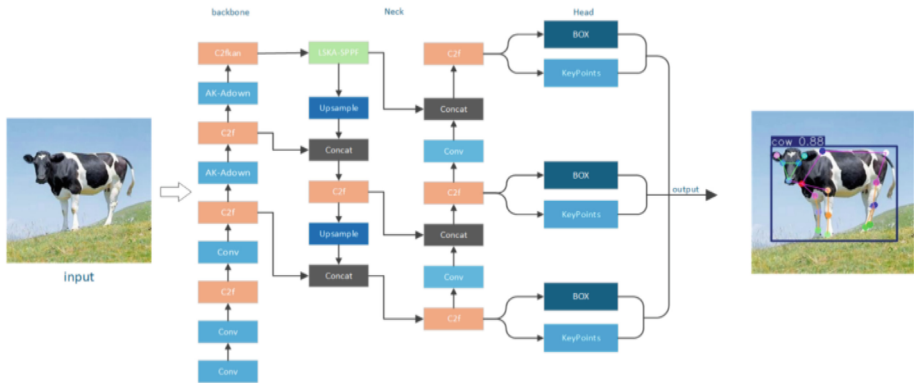


Fig. 2. The structure diagram of the improved YOLOv8-Pose model: Input an image to obtain the result diagram of its target detection and pose estimation.

3 Experiments and Analysis

3.1 Experimental Data Preparation and Environment

Datasets. In the experimental section, we conducted experiments using publicly available animal-pose datasets. These datasets are specifically designed for animal pose estimation tasks and include diverse poses (such as standing and running), significant occlusion issues, and considerable morphological differences among animals of the same species (e.g., different dog breeds). The animal-pose dataset [17] is highly diverse, features complex backgrounds, and provides consistent annotations. This approach not only fills the research gap in nonhuman pose estimation but also advances the development of cross-species pose analysis models. To further verify the effectiveness of our model, we conducted the same experiments on another publicly available dataset, AP-10K [18].

Benchmark Model and Experimental Preprocessing. This study employs YOLOv8-Pose as the baseline model for experimental improvements. To ensure data format consistency, the original datasets are processed as follows: the annotation files of the Animal-Pose datasets (20 keypoints) and the AP-10K datasets (17 keypoints) are converted into TXT files in YOLO format, where keypoint visibility is represented by 0 (invisible), 1 (occluded), and 2 (visible). For dataset partitioning, a stratified sampling method is adopted, dividing the data into training, validation, and test sets at a ratio of 7:2:1.

Experimental Environment. This paper is based on the Windows11 system. In terms of hardware configuration, an RTX4060 graphics card and Intel Core i7 processor are used. In the hyperparameter setting, the parameters are as follows: batch is set to 16, workers to 8, imgsz to 640, patience to 100, fraction to 1.0, mask_ratio to 4, max_det to 300, and plots to True. During the training process, the model was trained for 220 rounds in total, of which the last 10 rounds were processed with mosaic blur.

Evaluation Metrics. To thoroughly assess the proposed model’s effectiveness in animal pose estimation, we employ the YOLOv8-Pose evaluation framework. This evaluation considers both object detection and keypoint localization capabilities. In particular, we emphasize detection accuracy, using mAP@50, precision, and recall as core metrics to provide a comprehensive and reliable evaluation of the model’s performance.

3.2 Ablation Experiment

To validate the effectiveness of the LSKA module, we conducted systematic ablation experiments based on the YOLOv8-Pose baseline model, focusing on the following two aspects: First, to verify the efficacy of the improved LSKA module, we performed comparative experiments between the original YOLOv8Pose model and its enhanced version (Table 1).

Table 1. Comparison of LSKA before and after improvement with baseline.

Models	Params(M)	GFLOPs	Box			Pose		
			P	R	mAP50	P	R	mAP50
Yolov8n-pose	3357083	9.5	79.7	76.2	81.6	69.9	62.2	61.3
LSKA	3629979	9.7	81.6	71.2	81.7	72.6	59.6	59.4
Improve LSKA	3631003	9.7	81.3	77.3	84.4	82.5	61.2	67.5

The experimental results demonstrate that the improved LSKA module has excellent computational efficiency, with only a 0.8% increase in parameters and stable GFLOPs, confirming the effectiveness of its optimization. In the pose estimation task, the improved model shows significant advantages: a 6.2 percentage point improvement in mAP50, accompanied by a 12.6% increase in precision, despite a slight 1.0% decrease in recall. This performance shift indicates a substantial breakthrough in reducing false detections.

Table 2. Influence of different K_Sizes on the model performance.

k_size	Params(M)	GFLOPs	Box mAP50	Pose mAP50	FPS
7	3631003	9.7	84.4	67.5	26
11	3633051	9.7	85.1	67.9	22
23	3637147	9.7	83.2	65.4	19
35	3641243	9.7	84.9	66.4	17
53	3652947	9.7	85.5	65.2	14

To further investigate the impact of k_size on the multiscale feature extraction capability of LSKA, we conducted extended experiments comparing the effects of different convolutional kernel sizes (k_size = 7, 11, 23, etc....). The experimental results are presented in the above table.

As shown in Table 2, when `k_size` increases from 7 to 53, the number of model parameters (Params) tends to monotonically increase (3.63 M $\rightarrow$ 3.65 M), whereas the computational complexity (GFLOPs) remains stable (9.7). In terms of detection performance, the box mAP50 fluctuates with increasing `k_size`: it peaks at 85.1% for `k_size` = 11, decreases to 83.2% for `k_size` = 23, but rebounds to 85.5% for `k_size` = 53. This suggests that both medium-sized kernels (11) and very large kernels (53) may capture features beneficial for object detection, albeit through different mechanisms.

For pose estimation tasks, Pose mAP50 tends to decrease overall (67.5% $\rightarrow$ 65.2%), indicating that excessively large convolutional kernels may impair keypoint localization accuracy. Notably, the inference speed (FPS) significantly decreases as `k_size` increases (26 $\rightarrow$ 14), with a pronounced inflection point observed at `k_size` $\geq$ 23. This highlights the need to balance receptive field expansion with real-time performance requirements in practical applications.

3.3 Comparison of Experimental Results

This study takes YOLOv8-Pose as the baseline model and conducts systematic comparative experiments to evaluate the performance of pose estimation algorithms such as YOLOv8n-pose, YOLOv9tiny-pose, YOLOv10n-pose, and YOLO11n-pose.

Table 3. Comparison of experimental results.

Models	Params(M)	GFLOPs	Box			Pose		
			P	R	mAP50	P	R	mAP50
Yolov8n-pose	3357083	9.5	79.7	76.2	81.6	69.9	62.2	61.3
Yolov9tiny-pose	2235819	8.9	81.6	71.2	81.7	72.6	59.6	59.4
Yolov10n-pose	3046826	9.6	78.5	65.2	74.7	68.5	54.0	54.8
Yolov11n-pose	2933587	7.7	81.7	70.8	81.0	73.5	61.1	60.9
Ours	3631003	9.7	81.3	77.3	84.4	82.5	61.2	67.5

As shown in Table 3, from the two task dimensions of object detection and pose estimation, key performance indicators such as precision (P), recall (R), and mAP50 are selected for quantitative analysis.

The experimental results show that the improved YOLOv8-Pose model outperforms the original model in terms of the evaluation metrics. For object detection, the mAP50 is 2.8% higher than that of the original method, and for pose estimation, it improves by 6.2%. Additionally, we compared data for five animal species (cat, dog, cow, horse, and sheep). The data are presented in Table 4 below.

Table 4. Comparison of the experiment data of cats, dogs, cows, horses, and sheep in the animal pose datasets.

Animal	Metric	Model				
		Yolov8n	Yolov9t	Yolov10n	Yolo11n	Ours
Dog	Box	78.6	77.9	75.4	81.6	81.8
	mAP50	87.8	81.0	79.3	86.0	87.8
	Pose	57.4	65.3	58.2	66.2	74.8
	mAP50	52.9	43.6	44.5	56.3	62.7
Cat	Box	82.5	82.8	79.7	88	86.8
	mAP50	90.3	92.3	81.6	91.0	92.8
	Pose	68.8	74.6	70.4	71.3	84.4
	mAP50	66.9	60.4	59	61.2	75.8
Sheep	Box	66.4	62.1	70.5	79.5	69.8
	mAP50	68	61.8	65.4	70.7	71.5
	Pose	59.1	67	62.2	80.6	77.5
	mAP50	49	43.9	47.7	64.8	57.6
Horse	Box	86.2	77.7	93.3	80.5	84.5
	mAP50	95.2	82.5	89.3	81	96.1
	Pose	80.6	82.7	82.5	73.1	83.8
	mAP50	78.8	65.5	71.8	59.6	76.1
Cow	Box	84.0	63.5	73.6	78.8	84.0
	mAP50	66.8	63.8	57.7	75.5	73.9
	Pose	83.8	75.2	69.2	76.4	92
	mAP50	59.1	47.7	51.1	63.5	65.3

Comprehensive Performance Improvement: The proposed method (Ours) has significant advantages in both object detection and pose estimation tasks across five major animal categories (dog, cat, cow, horse, and sheep). Notably, it outperforms the baseline model YOLOv8n in terms of pose mAP50 by +9.8% (dog), +8.9% (cat), and +6.2% (cow). These results strongly validate the effectiveness of the LSKA module in modelling spatial relationships between keypoints.

Cross-Species Feature Analysis: For quadruped animals with complex joint structures (dog, cat, cow), our method shows particularly outstanding performance in pose estimation tasks, achieving Pose mAP50 scores of 62.7%, 75.8%, and 65.3%, respectively. This finding indicates that the LSKA module can effectively model anatomical constraints. Moreover, the model maintains stable performance in object detection tasks, with the box mAP50 exceeding 70% for all species. The detection accuracies for horses (96.1%) and cats (92.8%) are especially notable, highlighting the model’s robust feature extraction capabilities.

Although certain models may outperform specific metrics, our method achieves more balanced performance across both detection and pose estimation tasks. This suggests that the integration of the LSKA and SPPF modules contributes to overall feature enhancement rather than task-specific overoptimization.

To further validate the performance of the model and comprehensively evaluate its adaptability and robustness in complex real-world scenarios, we conducted the same experiments on another widely used public dataset, AP-10k. The AP-10k dataset contains diverse real-world scene samples, which can better reflect the model’s performance in authentic environments. Through experiments on this dataset, we were able to analyze the model’s generalization capabilities more thoroughly and verify its stability and accuracy across different scenarios. The experimental results are shown in Fig. 3.

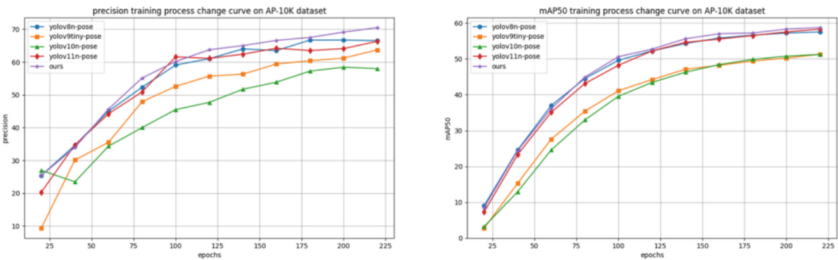


Fig. 3. On the left is the precision training process change curve, and on the right is the mAP50 training process change curve.

Our model achieves state-of-the-art performance on the AP-10K dataset. Both the precision and mAP50 curves consistently improve with training epochs, ultimately reaching peak values that demonstrate superior convergence and performance.

Table 5. The results of different models on the AP-10K dataset.

Model	Params(M)	GFLOPs	Input size	AP
vgg-16	19.1	16.3	192 × 256	66.3
ResNet-101	53.1	9.2	192 × 256	67.2
SimpleBaseline	53.0	12.1	256 × 256	68.1
VitPose-S	24.3	5.4	192 × 256	67.6
HRFormer	43.3	13.3	192 × 256	70.3
Ours	3.6	9.5	640 × 640	71.6

We systematically assessed the performance of several mainstream pose estimation models on the AP-10K dataset. Table 5 presents a comparison of keypoint detection accuracy (primarily AP) and computational efficiency metrics, including parameter count and FLOPs, across various input resolutions.

The experimental results show that this model significantly improves the detection accuracy while maintaining a low number of parameters and computational

complexity. This advantage enables it to demonstrate stronger competitiveness in a resource-constrained environment.

3.4 Visualization of Experimental Results

Figure 4 presents a systematic visual analysis of bovine pose estimation performance under four conditions: (a) high-resolution head close-ups, (b) complete body posture reconstruction, (c) multi-instance crowded scenes, and (d) partial occlusion scenarios.



Fig. 4. Visual Comparative Analysis of Cattle Instance Detection Based on Five Detection Models.

In terms of object detection, the figure illustrates the performance of different models in detecting cattle targets. For single-cattle scenarios with head close-ups (a) and full-body poses (b), all the models demonstrate accurate bounding box localization. However, in multicattle scenarios (c), discrepancies emerge among the models. Some exhibit reduced detection accuracy and completeness under dense target conditions, occasionally missing detections or generating false positives. In contrast, our model maintains more stable performance, reliably detecting all the targets with tighter bounding box fits.

For pose estimation, all the models roughly capture keypoints in head close-ups (a), although with varying precision in terms of angle and position. In full-body poses (b), the YOLO series models exhibit deviations in estimating limb and body contours, whereas our model more precisely reconstructs the cattle's posture, with skeletal lines closely aligned to the ground truth. The challenge intensifies in multicattle (c) and occluded (d) scenarios, where pose estimation errors increase significantly for YOLO-based models. Our model, however, has relatively high accuracy, highlighting its enhanced ability to capture and infer pose information in complex environments.

Combining the insights of object detection and pose estimation, our model outperforms other YOLO-Pose models, especially in challenging scenarios (multiple objects and occlusion conditions). This method has high positioning accuracy and key point accuracy, as well as strong robustness and generalizability. These advantages emphasize its practical value in real-world animal detection and attitude analysis applications.

4 Conclusion

We propose an enhanced YOLOv8n-Pose model, built upon the original architecture, for animal pose estimation. To improve the fusion of multiscale features, we integrate an optimized LSKA mechanism with the SPPF module. Due to the complexity and variability of animal postures, conventional convolutional neural networks often struggle to extract discriminative features from animals with diverse sizes and shapes. The incorporation of separable convolutions in the LSKA module enables more refined processing of input features, thereby enhancing the model's ability to capture fine-grained spatial details. Although there was a slight improvement in animal pose estimation in this study, some obvious limitations remain. For example, the model's detection accuracy varies significantly across different species. In this work, while the improved YOLOv8-Pose model significantly improved the evaluation metrics for sheep, cats, and dogs, a slight decrease in performance was observed for horses and cows. These experimental results highlight the differences between animal species, which may lead to the model's inability to handle all animals equally well. Future research can explore advanced data augmentation strategies and transfer learning techniques to mitigate the limitations caused by the lack of sufficient animal pose datasets. Data augmentation can help enrich the dataset, whereas transfer learning can enable the application of advanced techniques from human pose estimation to animal pose estimation, thereby improving the model's performance in unseen environments. Additionally, real-time performance is a critical factor in practical applications. Although the proposed model demonstrates higher accuracy, further optimization of computational efficiency is still needed for real-time detection and pose estimation tasks.

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